

PP Lecture 4 PDC

Part 1: Past Papers

QUIZ 01 - FALL 2022 / SPRING 2023 • QUESTION # 3

Q: Most modern day parallel architectures would fall under which category of Flynn's taxonomy?

- a. SISD**
- b. MISD**
- c. SIMD**
- d. MIMD**

A: d. MIMD

MIDTERM 1 - SPRING 2022 • QUESTION # 1(I)

Q: Single Program with Multiple Data (SPMD) is a close variant of _____ in Flynn taxonomy:

- a. SIMD**
- b. MIMD**
- c. MISD**
- d. SISD**

A: b. MIMD

MIDTERM 2 - SPRING 2023 • QUESTION # 1(A)(I)

Q: GPU is an example of:

- a. Shared Memory SIMD**
- b. Distributed Memory SIMD**
- c. Shared Memory MIMD**
- d. Distributed Memory MIMD**

A: a. Shared Memory SIMD

SESSIONAL-I / MIDTERM 1 - SPRING 2024 • QUESTION # 1(I)

Q: How we can classify the type of computer architecture on a typical, consumer-grade multi-core laptop computer released in the year 2024 out of the following?

- A) SISD
- B) SIMD
- C) MIMD
- D) MISD

A: Correct answer: (A), (B), and (C) — because using a single core acts as SISD, using all cores concurrently acts as MIMD, and modern CPUs include vector instruction sets, incorporating SIMD.

QUIZ 01 - SPRING 2025 • QUESTION # 2

Q: Name a hardware for each of the following four, which can be classified as:

- (a) SISD
- (b) SIMD
- (c) MIMD
- (d) MISD

A:

- **(a) SISD (Single Instruction Stream, Single Data Stream):** Pentium Pro processor with just one core
- **(b) SIMD (Single Instruction Stream, Multiple Data Stream):** Vector processors, SSE / AVX units in Intel processors
- **(c) MIMD (Multiple Instruction Streams, Multiple Data Streams):** Multi-core processors
- **(d) MISD (Multiple Instruction Streams, Single Data Stream):** No practical hardware exists; systolic arrays (found in Google's TPU) and specialized NASA shuttle hardware are sometimes cited.

SESSIONAL-I / MIDTERM 1 - SPRING 2025 • QUESTION # 3A

Q: You are designing a cloud-based image processing service where each uploaded image is independently transformed (e.g., resized and enhanced). With thousands of concurrent uploads, which Flynn's Taxonomy architecture best supports parallelism and scalability? (Just name it.) Provide rationale for your answer in a line.

A: Multiple answers are acceptable:

- **MIMD:** Multiple independent processors work on independent images for maximum performance and scalability.
- **SPMD or SIMD:** Since Flynn's taxonomy is flexible, it resembles SIMD because all processing elements run the same transformation operation on independent data streams.

Part 2: Book Questions

INTRODUCTION TO PARALLEL COMPUTING, 2ND EDITION - ANANTH GRAMA CHAPTER # 9 • PROBLEM # 9.1

Q: Consider a compare-split method for sorting blocks of elements. Is this method better suited for MIMD or SIMD parallel computers?

A: It is better suited for MIMD parallel computers. MIMD systems allow independent control flow and asynchronous processing per processor, whereas SIMD systems incur significant overhead when conditional execution forces processors into idle states during block comparisons.